AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

MODEL EXAMINATION PAPER 2026 AND ONWARDS

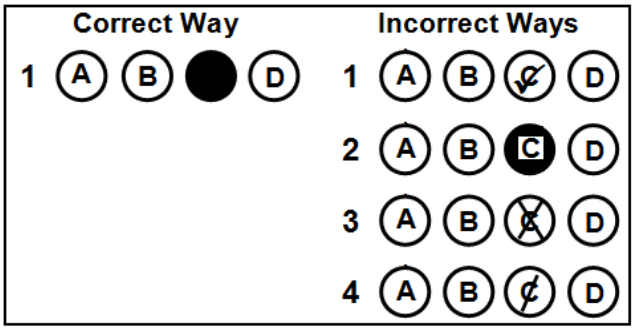
Physics Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS:

1. Check your name and school information on the question paper and Optical Mark Recognition (OMR) sheet. Sign on both if the information is correct.
2. Read each question carefully.
3. Answer all questions on the provided OMR sheet within the allotted time. DO NOT mark your answers on the question paper.
4. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
5. Each question has four choices: A, B, C, and D. Choose ONE correct answer only. On the OMR sheet, completely fill in the circle corresponding to your choice with a pencil as shown in the example below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with an eraser, before filling in the new circle. Ensure your marks are dark and clear. Do not make stray marks or leave incomplete fillings as the OMR machine ONLY records what is in the circles.
7. Do not fold, tear, or damage the OMR sheet. Damaged sheets may be scanned incorrectly.
8. You may use a scientific calculator if you wish.

1. The dimensions of momentum are represented as

   A. $[ML^2T^{-1}]$
   B. $[MLT^{-1}]$
   C. $[MLT^{-2}]$
   D. $[ML^{-1}T]$

2. Uncertainty in measurement can occur due to the

   I. limitation of instruments
   II. natural variations
   III. frequent errors

   A. I only.
   B. III only.
   C. I and II.
   D. II and III.

3. If the dot product of two vectors (**X** and **Y**) is zero, then which of the following conditions is TRUE?

   A. Both vectors are parallel to each other
   B. Both vectors are opposite to each other
   C. Either of the two vectors is a unit vector
   D. Either of the two vectors is a null vector

4. The cross product of two vectors (**X** and **Y**) is a negative vector when the angle between them is

   A. 0°
   B. 90°
   C. 180°
   D. 270°

5. The law of conservation that is used in the propulsion of rockets is

   A. heat.
   B. mass.
   C. energy.
   D. momentum.

6. A moving car crashes into a stack of sandbags. It hits the sandbags and passes through them.

   When the car hits the sandbags, the total momentum of the car

   A. increases.
   B. decreases.
   C. becomes zero.
   D. remains the same.

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7. An object falls freely from rest. The total distance covered by it in 2 s will be

(**Note**: The acceleration due to gravity 'g' is 9.8 m/s^2.)

A. 9.8 m.
B. 19.6 m.
C. 29.4 m.
D. 39.2 m.

8. A lighter metallic ball (m_1) collides with a much heavier metallic ball (m_2) which is at rest, and ($m_1 << m_2$), then after an elastic collision, the heavier ball (m_2) will

A. remain at rest.
B. gain the lighter ball's velocity.
C. move in the opposite direction.
D. move with double the velocity of the lighter ball.

9. In a conservative field, the work done in moving a body around a closed loop is always

A. zero.
B. negative.
C. dependent on the shape of the loop.
D. dependent on the length of the path.

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10. A moving truck applies brakes and stops after traveling 20 m. If its kinetic energy was 4000 J, then the average braking force will be

A. 200 N.
B. 3980 N.
C. 4020 N.
D. 80000 N.

11. The given figure shows the first attempt of a long jump of an athlete in Olympics.



Which of the following factors of the athlete will MAINLY help him to improve his own record in the next attempt?

A. Angle of projection of the athlete
B. Acceleration of the athlete
C. Body shape of the athlete
D. Weight of the athlete

12. Missiles **I**, **II**, **III** and **IV** are launched at 35°, 45°, 55° and 65°, respectively from the launching pad with same initial velocity.

    The shortest distance will be covered by

    A. Missile I.
    B. Missile II.
    C. Missile III.
    D. Missile IV.

13. When an object moves with uniform speed in a circular orbit, its centripetal acceleration will be directed

    A. along the circumference of the circle.
    B. along the direction of motion of the object.
    C. towards the centre of the circle along the radius.
    D. away from the centre of the circle along the radius.

14. In all of the following sports, the law of conservation of angular momentum is used EXCEPT in

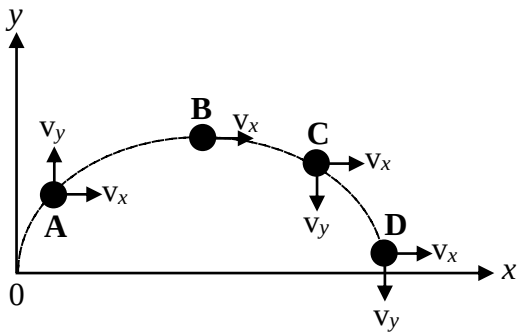
    A. ice-skating.
    B. gymnastics.
    C. horse riding.
    D. board diving.

15. The acceleration produced by virtue of changing direction of velocity of an object moving in a circular path is called

    A. average acceleration.
    B. tangential acceleration.
    C. centripetal acceleration.
    D. instantaneous acceleration.

16. The given diagram shows the projectile motion of a ball.

    The magnitude of the vertical component of velocity (v) will be maximum at point



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17. The direction of the angular acceleration of a body is always

    I. along the axis of rotation
    II. along the tangent to the path
    III. perpendicular to the axis of rotation

    A. I only.
    B. III only.
    C. I and II.
    D. II and III.

18. A copper wire of length 3 m and an area of 2 m^2 is stretched by a force of 500 N.

    If Young's modulus of the copper wire is 1.1×10^{11} Pa, then the elongation of the wire will be

    A. 6.8×10^{-9} m.
    B. 3.3×10^{8} m.
    C. 2.2×10^{14} m.
    D. 1.4×10^{-14} m.

19. All of the following are applications of Bernoulli's effect EXCEPT

    A. aerofoil.
    B. atomizer.
    C. venturi meter.
    D. vacuum cleaner.

20. If each particle of a fluid passing through a point follows the same path, then this flow is called

    A. normal flow.
    B. irregular flow.
    C. turbulent flow.
    D. streamline flow.

21. According to the equation of continuity, if the cross-sectional area of a pipe decreases, then the speed of the fluid passing through it

    A. increases.
    B. decreases.
    C. remains the same.
    D. varies unpredictably.

22. The statement which is CORRECT with respect to the Bernoulli's theorem is that the

    A. speed of fluids is low at low pressure.
    B. speed of fluids is high at low pressure.
    C. pressure of a fluid is independent of its speed.
    D. theorem is valid only for turbulent flow of a liquid.

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23. A mass ‘m’ is hanging on an inextensible string to make a simple pendulum of time period ‘T’.

If the mass becomes ‘3 m’, then the time period of the simple pendulum will be

A. $\frac{T}{3}$.
B. T.
C. 3T.
D. $\sqrt{3}\,T$.

24. A particle is moving in a circle with uniform speed, for which the projection of the particle is moving along the diameter of the circle.

In the given situation, the motion of projection is

A. linear.
B. rotatory.
C. circulatory.
D. simple harmonic.

25. Resonance is a phenomenon in which a vibrating system or external force drives another system to oscillate with greater amplitude at specific frequencies.

This phenomenon helps in determining the

A. forced vibration.
B. natural frequency.
C. energy dissipation.
D. amplitude of a vibrating body.

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26. Loud music produced by beating a wooden drum is an example of

A. beats.
B. free vibrations.
C. forced vibrations.
D. damped oscillations.

27. When a car moves on any uneven surface, then its shock absorbers play an important role.

The working of shock absorbers exemplifies

A. free oscillation.
B. force oscillation.
C. damped oscillation.
D. undamped oscillation.

28. When the bob of a simple pendulum is at the mean position during oscillation, its kinetic energy will

    A. be zero.
    B. be minimum.
    C. be maximum.
    D. remain constant.

29. If a 0.5 kg mass is attached to a spring constant of 200 N/m, then the time period of oscillation will be

    A. 0.05 s.
    B. 0.314 s.
    C. 20.0 s.
    D. 125.6 s

30. The option which is TRUE about the speed of wave on a string is that speed depends on the

    A. frequency.
    B. amplitude.
    C. mass per unit length of the string.
    D. atmospheric conditions of the surrounding.

31. If a wave has a frequency of 10 Hz, then the time period of the wave will be

    A. 0.1 s.
    B. 1.0 s.
    C. 10 s.
    D. 100 s.

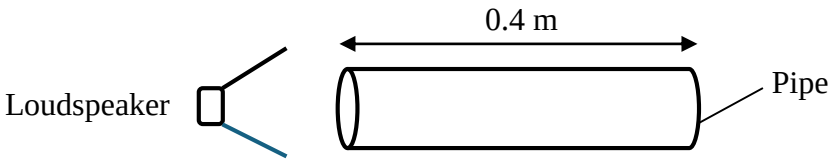
32. When two tuning forks of nearly the same frequencies are sounded together, a note of alternately increasing and decreasing intensity will be heard.

    Based on the given description, the note is called

    A. beat.
    B. waveform.
    C. diffraction.
    D. polarisation.

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33. One end closed pipe has a loudspeaker at its open end. A stationary sound wave is formed in the air within the pipe with an antinode at the open end of it, as shown in the given figure.



(**Note**: The speed of sound in air is 343 m/s.)

The frequency of sound that produces a stationary wave from the loudspeaker is

A. 214.4 Hz.
B. 285.8 Hz.
C. 428.8 Hz.
D. 857.5 Hz.

34. When a listener moves away from a stationary source of sound, then the apparent change in the pitch and frequency of sound for the listener is

	Pitch	**Frequency**
A	low	low
B	high	low
C	low	high
D	high	high

35. The phenomenon of wave that distinguishes between transverse and longitudinal waves is

A. refraction.
B. diffraction.
C. interference.
D. polarisation.

36. The resultant displacement at any point due to the sum of displacements of two or more waves is called

A. interference.
B. Doppler's effect.
C. Huygen's principle.
D. principle of superposition.

37. If the analyser is rotated from 0° to 90°, then the transmitted intensity of polarised light will

A. decrease to zero.
B. remain the same.
C. vary unpredictably.
D. increase to maximum.

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38. Which of the following is NOT the application of polarisation of light?

A. Photographers use filters to reduce the glare in order to get clear pictures.
B. Drivers use sunglasses that diminish the brightness of headlights of a car.
C. Astronomers use lenses in telescopes to enlarge the image of celestial bodies.
D. Fishermen use eye-wear which decreases the reflection of light from the water.

39. The optical media that is used to eliminate the glare of light produced by the road surface during driving is

A. sun screen.
B. convex lens.
C. plane mirror.
D. polaroid glass.

40. Colours seen from an oil film on water depends on the following factors EXCEPT

A. angle of view.
B. density of the oil.
C. index of refraction.
D. thickness of the film.

41. According to Huygen's principle, light travels in the form of

I. photons
II. corpuscles
III. wavefronts

A. I only.
B. III only.
C. I and II.
D. II and III.

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42. In Young's double slits experiment, the relation used to find the position of the dark fringes is

(**Note**: All the symbols have their usual meaning.)

A. $Y = (m + 1) \lambda L / d$
B. $Y = (m - 1) \lambda L / d$
C. $Y = (m + 1/2) \lambda L / d$
D. $Y = (m - 1/2) \lambda L / d$

43. Michelson's interferometer is used to measure

A. velocity of light.
B. amplitude of light.
C. frequency of light.
D. wavelength of light.

44. All Carnot engines have zero efficiency if they are operating between two bodies with

A. the same temperature and the same nature of the working substance.
B. different temperatures and the same nature of the working substance.
C. the same temperature irrespective of the nature of the working substance.
D. different temperatures irrespective of the nature of the working substance.

45. The CORRECT sequence of the working principle of Carnot engine is

A. Adiabatic compression $\rightarrow$ isothermal expansion $\rightarrow$ isobaric compression $\rightarrow$ isochoric expansion
B. Isothermal expansion $\rightarrow$ isobaric compression $\rightarrow$ adiabatic compression $\rightarrow$ isochoric expansion
C. Isothermal expansion $\rightarrow$ adiabatic expansion $\rightarrow$ isobaric compression $\rightarrow$ adiabatic compression
D. Isobaric compression $\rightarrow$ isothermal expansion $\rightarrow$ isochoric expansion $\rightarrow$ adiabatic compression

46. According to Boyle's law, volume is inversely proportional to pressure of a gas at constant

I. mass
II. density
III. temperature

A. I only.
B. II only.
C. I and III.
D. II and III.

47. A thermodynamics process in which an ice cube starts melting and converts into water droplets while keeping the temperature of its surrounding constant is called an

A. isobaric process.
B. isochoric process.
C. adiabatic process.
D. isothermal process.

48. According to the 2nd law of thermodynamics, heat can be converted into mechanical work if the system contains

A. two heat reservoirs at the same temperature.
B. two heat reservoirs at different temperatures.
C. an engine and a heat reservoir at the same temperature.
D. an engine and two heat reservoirs at different temperatures.

49. In thermodynamics, when any two objects are rubbed together, then their internal energy will

A. increase.
B. decrease.
C. become zero.
D. remain constant.

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50. All of the following are the examples of reversible process EXCEPT

A. boiling of raw eggs.
B. compression of a helical spring.
C. slow compression of oxygen gas.
D. freezing of a carbonated drink into ice.

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END OF PAPER

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